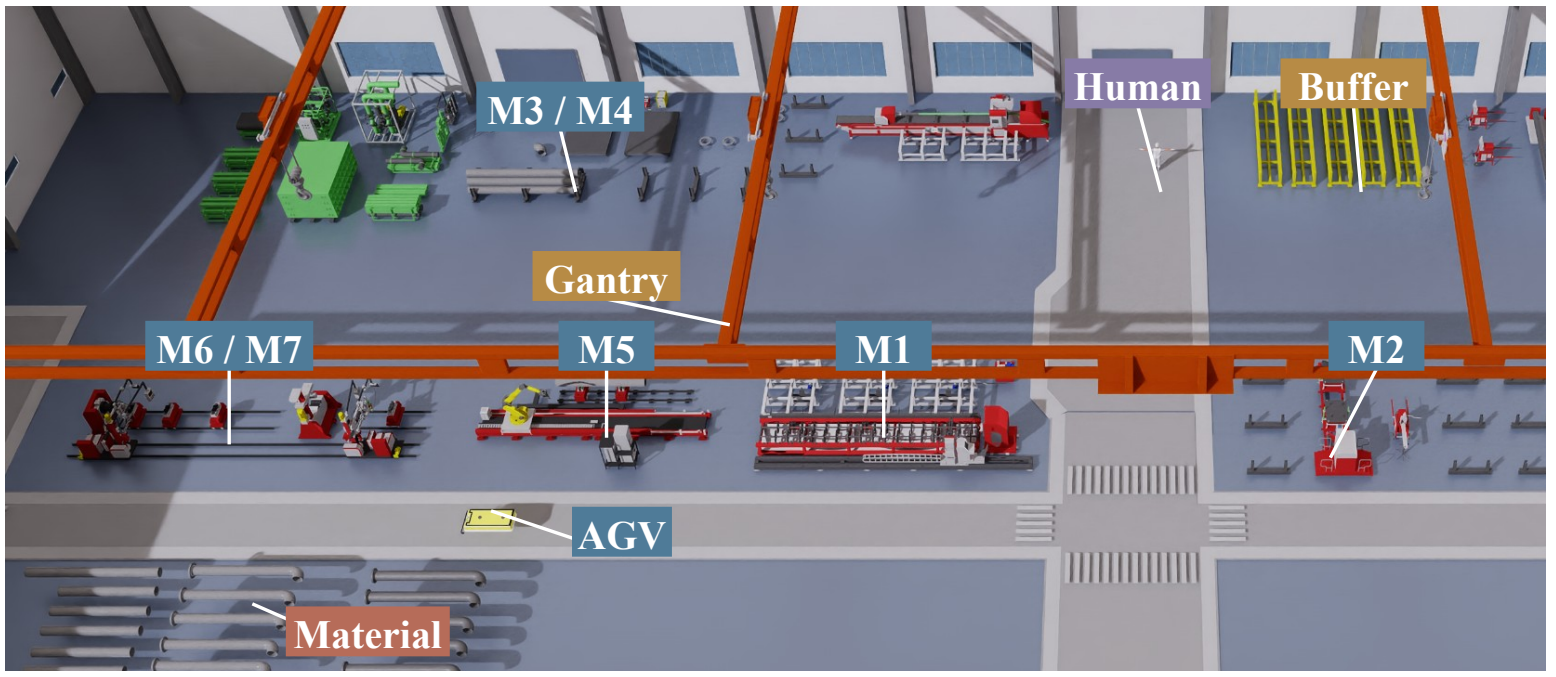
(a) Digital-twin production environment and resource entities



Machine / workstation identifiers

- **M1** CNC pipe intersection cutting machine
- **M2** Large pipe roll-grooving machine
- **M3** Workbench 0
- **M4** Workbench 1
- **M5** Robotic root-welding station
- **M6** Rotary surface-welding station 0
- **M7** Rotary surface-welding station 1

M3 / M4 support spot welding and painting.
AGV: automated guided vehicle.

(b) Manufacturing stages and shared material-handling subtasks

Manufacturing stages

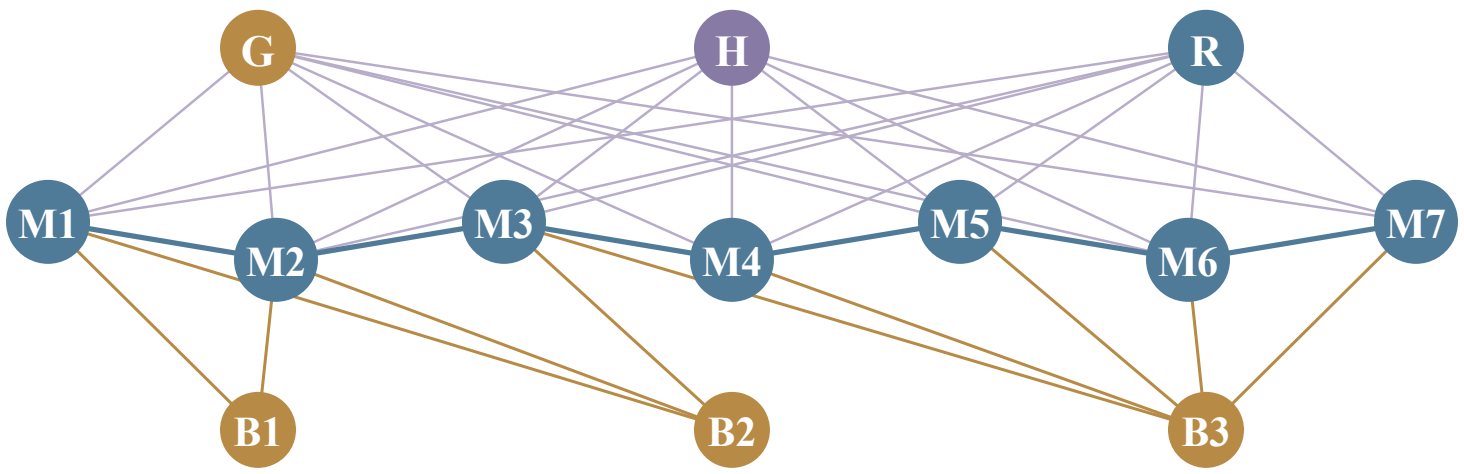
1. **Pipe cutting at M1**
   Raw pipe delivered; worker operates the cutting machine.
   Cut pipe is released for transfer.
2. **Roll grooving at M2**
   Cut pipe transferred; worker operates the grooving machine.
   Grooved pipe proceeds to assembly.
3. **Assembly and spot welding at M3 / M4**
   Pipe, flange and elbow supplied to a selected workbench.
   Assembly and spot welding yield a semi-finished pipe.
4. **Argon-arc root welding at M5**
   Semi-finished pipe delivered to the welding robot.
   Worker operates the station for root welding.
5. **MIG surface welding at M6 / M7**
   Root-welded pipe transferred to a rotary welding station.
   Surface welding prepares the pipe for painting.
6. **Rust-proof painting at M3 / M4**
   Welded pipe returned to a workbench for rust-proof treatment.
   The finished water pipe is ready.

Shared handling subtasks

7. **Processing and outbound handling**
   Worker approaches the station
   → Controls the machine during processing
   → Waits for a free gantry
   → Assists loading for outbound transfer
8. **Transfer within one gantry zone**
   Worker and gantry approach the material
   → Load onto gantry and carry to destination
   → Place at the station or storage area
9. **Transfer across gantry zones**
   Source gantry loads an AGV and is released
   → AGV travels to the destination
   → Free destination gantry unloads and places

(c) Multi-entity structural graph

G: gantries (4) H: workers (5) R: AGVs (4)



B1: BlackStorage B2: GroundStorage B3: YellowStorage

Structural relations

- Workstation backbone
- Human / transport–station links
- Buffer–station affinities

38 nodes: 7 workstations, 4 gantries, 5 workers, 4 AGVs and 18 buffers.

Undirected associations, with representative G / H / R and buffer nodes. Repeated nodes, self-loops and within-type transport links are omitted.

Material availability enters resource features through supply, waiting and buffer occupancy.